

Md Kaif Imteyaz

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE student with research internships at Michigan State University and under an Amazon economist, specializing in LLM fine-tuning, retrieval-augmented generation, and encoder-level model behavior across BART, FLAN-T5, GPT-2, and Qwen-2 architectures. Experienced in reading and modifying transformer model internals via PyTorch and designing end-to-end retrieval pipelines. Co-author of an LLM benchmarking paper (work in progress); ranked 2nd at the AMD AI Reinforcement Learning Hackathon (200+ teams) and 7th at AISEHack 2026 (2,000+ participants).

EDUCATION

Jamia Hamdard	New Delhi, India
Bachelor of Technology in Computer Science & Engineering; 8.53 CGPA (avg. till 7 th sem)	2023 – 2026
Jamia Millia Islamia	New Delhi, India
Diploma in Computer Engineering; 83.38%	2020 – 2023

TECHNICAL SKILLS

Programming: Python, JavaScript, Java
LLM & NLP: Qwen, FLAN-T5, BART, GPT-2, Hugging Face, Transformers, LoRA, QLoRA, RLHF, RAG, Prompt Engineering, OpenAI APIs
AI/ML: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Weights & Biases (wandb), Model Fine-tuning, Transfer Learning, Model Quantization, Object Detection, Data Preprocessing
Web & Backend: Node.js, Express.js, React.js, Next.js, Tailwind CSS, REST APIs, FastAPI
Databases: PostgreSQL, MySQL, Supabase
Tools: Git, Postman, Figma, Google Colab, Jupyter Notebook

WORKING PAPERS

Detecting Design Theater in Reasoning Models | LLM Benchmarking | (*Work in Progress*)

Co-Authors: Kashif Imteyaz, Daniel Lee, Md Kaif Imteyaz, Saiph Savage

Developed a benchmark for detecting when a language model generates convincing rationales for features it did not actually implement, quantifying the authenticity gap in reasoning model outputs across creative tasks. Involves systematic evaluation of model internals and reasoning trace analysis.

CATS: Culturally Adaptive Therapeutic System | *Final Undergraduate Project* | (*Work in Progress*)

Designed an agentic system for physiotherapists to generate personalized exercise videos aligned with patients' cultural backgrounds through culturally appropriate demonstrations.

EXPERIENCE

Data Science Intern Sept. 2025 – Feb. 2026

Worked with Prof. Aaditya Dar, Economist at Amazon

- Implemented a scalable data pipeline that cleaned, standardized, and structured approximately 2 million government asset declaration records for downstream use.
- Curated a 1,000-row labeled dataset and studied behavioral differences across a suite of language model architectures (BART-base, FLAN-T5, GPT-2, Qwen-2) to automate attribute extraction through targeted fine-tuning.
- Designed a model selection and pipeline architecture for entity extraction and record linkage across multi-year personnel data, reasoning about encoder-level tradeoffs between model depth, generalization, and throughput.

Research Intern Mar. 2025 – Aug. 2025

Michigan State University (worked with Prof. Prabhat Barnwal)

- Developed the Peer Knowledge Exchange Platform to improve retrieval speed, accessibility, and user experience for government officers in Punjab, India.
- Deployed a WhatsApp chatbot, Sewa Assist, enabling officers to locate and access program resources faster through conversational retrieval.

- Integrated an on-site chatbot that reduced user friction and improved resource discovery through keyword-driven retrieval logic.
- Delivered the solution using Next.js with version control via GitHub for long-term maintainability and scalability.

Full Stack Developer Intern

Dec. 2022 – Jan. 2023

FTK Center for Information Technology, Jamia Millia Islamia

New Delhi, India

- Led full stack development of an alumni community web application serving 2,000+ users, improving engagement using HTML, CSS, JavaScript, and PHP.
- Optimized site architecture for mobile responsiveness and reduced page load times by 20%.
- Collaborated in a four-member Agile team to ship three major features within six weeks.

PROJECTS

- De-Tail: Agentic AI Search Assistant** | *React, OpenAI API, Tavily API* | [Repository](#) 2025
- Engineered a retrieval-augmented generation pipeline that exposes its full step-by-step reasoning chain, integrating OpenAI Reasoner with Tavily API for real-time web retrieval.
 - Designed end-to-end retrieval flow from query intake through document fetching, reranking signal generation, and response synthesis, with full transparency at each stage.
- Ink2Info: Digitizing Health Informatics** | *Python, Next.js, FastAPI, OpenAI, Gemma 7B* | [Repository](#) 2026
- Developed a prescription digitization pipeline using OpenAI and Gemma 7B with a Fabric.js canvas for real-time handwriting extraction, directly modifying inference behavior of the underlying models for domain adaptation.
 - Implemented role-based access control with HMAC-signed QR codes for secure data sharing across multiple user roles.
- OceanDataSync: Oceanographic Data Pipeline** | *Python* | [Repository](#) 2025
- Developed an automated pipeline aggregating and standardizing large-scale oceanographic datasets from NOAA, Copernicus Marine, Argo, and NCEI.
 - Implemented chunked processing, YAML-based configuration, robust retry logic, and structured CSV and JSON logging for reproducibility.
- Anti Corrupt Infrastructure** | *React, Node.js, OpenRouter (OpenAI)* | [Repository](#) 2025
- Designed an AI-based transparency platform for infrastructure projects with automated fund tracking, project monitoring, and milestone verification.
- Vulture: Vulnerability Scanner** | *React, Node.js, OWASP ZAP API* | [Demo Video](#) 2023
- Developed a web application identifying security vulnerabilities using OWASP ZAP API during a 48-hour hackathon; ranked in the top 10 out of 100+ submissions.

ACHIEVEMENTS

- **2nd Place**, AMD AI Reinforcement Learning Hackathon among 200+ teams, organized by AMD, Yardi School of AI (IIT Delhi), and Unsloth AI (2026)
- **7th Rank** (out of 2,000+ participants, two-phase), AISEHack 2026 organized by ANRF in collaboration with IBM and IIT Delhi, hosted at IIIT Hyderabad (2026)
- **Top 10 / 100+ Teams**, Developed a vulnerability scanner using OWASP ZAP API in a 48-hour hackathon (2023)

CERTIFICATIONS

- Getting Started with Artificial Intelligence** July 2025
IBM SkillsBuild Online
- Data Fundamentals** July 2024
IBM SkillsBuild Online

EXTRACURRICULAR ACTIVITIES

- Attendee, AI Summer School** May 2024 & July 2023
IIIT Delhi New Delhi, India
- Core Member and Web Contributor** 2023 – 2024
Google Developer Groups, Jamia Hamdard New Delhi, India
- Volunteer Frontend Developer** Apr. 2024
International Conference on Sustainability and Digitalization, Jamia Hamdard New Delhi, India
- Member, Programming and Animation Team** 2022
E Cell JMI New Delhi, India